

Infrastructure State Report

Repository-derived overview of the commercial platform, research proof layer, developer tooling, security controls, and operational dependencies.

REPORT DATE	SCOPE
30 July 2026	Implemented repository surfaces and their stated release boundaries

Assessment basis

This report is based on the local Next.js application source, contracts, tests, and configuration reviewed on 30 July 2026. It is not a penetration test, financial audit, or independent measurement of production performance. Product outcome statements remain workload-specific and require customer benchmark acceptance criteria.

Operating model

Maha operates a dual-domain model: mahastrategies.com provides commercial product and API surfaces, while research.mahastrategies.com provides citations, structured claims, and mathematical context. The two are connected through research URLs, provenance references, and machine-readable metadata.

1. Executive state

The repository contains a broad, integrated platform rather than a single landing-page estate: research atlas modules, commercial optimization products, authenticated APIs, provenance tooling, context middleware, an MCP control layer, and developer-adoption assets are all present.

Area	State	Evidence / boundary
Commercial product surfaces	Implemented	Five product pages with calculators, research links, canonical metadata, and SoftwareApplication JSON-LD.
Research proof layer	Implemented	Atlas modules, structured source and claim records, static claim routes, and llms.txt manifest.
API and key platform	Implemented with active operational follow-up	Upstash-backed API keys, credit accounting, CORS, OpenAPI, SDK, CLI, and selected v1 endpoints.
Payments	Implemented contract	Stripe webhook verification, credit-pack mapping, and idempotent crediting are present; production configuration remains an operator responsibility.
Quality gates	Validated locally	Latest local checks: TypeScript compile passed; node test suite passed 161 tests. Deployment validation remains required after each production change.

Current release note

The latest key-service investigation identified and corrected a Redis HGETALL response-normalization issue. Production should be redeployed with that source change, then rechecked with a newly issued key. Temporary diagnostic logging should be disabled once the balance endpoint returns 200, because raw header dumps can expose internal platform metadata.

Architecture at a glance

Research proof layer	Commercial and developer layer	Operations and controls
Atlas modules Claim ledger and citations Static claim pages llms.txt manifest	Product pages and calculators OpenAPI v3.1 and v1 APIs SDK, CLI, provenance widget Context compiler and MCP surfaces	Vercel / Next.js 16 Upstash Redis credits and rate limits Stripe payment events OpenAI-compatible upstream proxy

2. Product portfolio and capability surfaces

Product	Implemented capability	Proof / interface
Maha Tensor-Opt	GPU-native QUBO / Ising optimization workflow with declared tensor-network parameters and benchmark framing.	research.mahastrategies.com/atlas/tensor-networks
Maha Geometric AI	Symmetry-aware neural-network positioning for geometry-bound physics and engineering work, with declared invariance assumptions.	research.mahastrategies.com/atlas/geometric-ai
Maha QEC-Compiler	Hardware-agnostic holographic QEC code-synthesis and fault-tolerant layout-planning surface.	research.mahastrategies.com/atlas/holographic-qec
Maha Landscape-Opt	Topology-aware high-dimensional constraint search for EDA, logistics, and multi-agent optimization.	research.mahastrategies.com/atlas/landscape-opt
Context Compiler	Deterministic, source-linked Context Packs with byte/token estimates and visible budget exclusions.	mahastrategies.com/context-compiler
Enterprise MCP Gateway	Tenant-scoped registered MCP server inventory, method/tool allowlists, and privacy-preserving audit records.	mahastrategies.com/enterprise-mcp-gateway

Commercial engagement boundaries

The commercial surfaces are deliberately bounded. Tensor-Opt is framed around declared acceptance tests rather than blanket quantum-advantage claims. Geometric AI validates declared constraints, not universal scientific correctness. QEC overhead reduction is a planning scenario, not a guarantee. Landscape-Opt reports residuals and workload-specific outcomes rather than claiming universal global optimality.

Interactive demand-generation surfaces

- Token Cost and Payload Estimator: local-first byte and approximate-token calculations, provider cost estimates, simulated compression scenarios, and API key CTA.
- Bulk Log ROI Auditor: browser parsing of JSONL / JSON traces with capped upload handling and planning-level enterprise ROI outputs.
- Per-product ROI calculators: Tensor-Opt, Geometric AI, Holographic QEC, and Landscape-Opt pages expose scenario planning rather than production performance promises.

3. Research, provenance, and discoverability

The research domain is designed to anchor commercial propositions in citable, structured material instead of unlinked landing-page copy.

Capability	Infrastructure status
Atlas ledgers	Typed source and claim modules for tensor networks, geometric AI, holographic QEC, and landscape optimization. Catalog and aggregate integrity checks connect modules.
Claim ingestion	scripts/expand-graph.ts ingests Markdown, LaTeX, and JSON from research-source into typed MPS claim data.
Claim publishing	Static /claims/[slug] pages use build-time parameters, citation sections, related-claim links, canonical metadata, and TechArticle / DefinedTerm JSON-LD.
Crawler manifest	/llms.txt dynamically provides active claim summaries and useful API endpoints in text form for AI crawlers.
Attribution widget	A zero-runtime-dependency public widget renders a provenance badge that links external content back to the claim URL.

SEO and schema posture

Commercial product pages use canonical paths and a single SoftwareApplication node with BusinessApplication and operating-system metadata. Research pages use TechArticle and DefinedTermSet / DefinedTerm relationships pointing to the organization and named-person identifiers. The intentional exclusion of unsold Product schema avoids invalid rich-result requirements.

Content quality safeguard

The ingestion pipeline stores claim status as VERIFIED, SOURCED, ILLUSTRATIVE, or UNVERIFIED. Related-claim linking and citations reduce orphaned, repetitive, or context-free pages. The remaining governance requirement is editorial review of every new source before publication; automation alone cannot establish scientific validity.

4. API platform, developer tooling, and AI middleware

Surface	Current capability
OpenAPI contracts	OpenAPI 3.1 registry covers optimization endpoints and documents request/response schemas.
Optimization APIs	v1 mock job contracts for Tensor-Opt, Geometric AI, Holographic QEC, and Landscape-Opt return accepted execution payloads and research citations.
Maha API keys	No-password key issuance with a credit ledger, tier metadata, rate limits, zero-data-retention flag, and a one-time raw-key response.
MahaClient SDK	Zero-dependency TypeScript client using native fetch, typed errors, and exponential retry for 429 responses across Node, Bun, Deno, browser, and Edge runtimes.
CLI	Audit local Markdown / TXT for estimated prompt-token savings and issue a key from the terminal; JSON output is designed for CI consumption.
OpenAI-compatible proxy	Non-streaming Chat Completions proxy accepts standard payloads, compacts eligible text history, forwards to OpenAI, and preserves upstream response bodies.

Context Pack controls

The Context Compiler measures original and compiled bytes, estimated tokens, duplicate removal, and source coverage. It returns source references and hashes; the product page states that original documents are not stored. The chat proxy only compacts text-only historical messages and preserves multimodal and tool-call messages verbatim to avoid changing their semantics. Streaming is explicitly not supported in the current endpoint.

MCP delivery boundary

The Enterprise MCP Gateway routes JSON MCP messages only to registered public HTTPS upstreams. It has tenant binding, explicit method/tool allowlists, and logs method/outcome/request hash rather than request bodies. It intentionally excludes private upstream access, upstream credential storage, bearer-token forwarding, browser-originated authorization, and SSE streaming in this release.

5. Security, billing, and operational dependencies

Control area	Implemented behavior
API authorization	Proxy requires a Bearer Maha API key for protected /api/v1 paths. Missing/invalid keys are 401; depleted keys are 402; unavailable authorization service is fail-closed 503.
Key management exclusions	Only key generation, balance, and checkout routes bypass the primary key guard. CORS preflight is handled for browser SDK use.
Redis implementation	Upstash REST URL and token are sanitized for stray quote/whitespace input. Key records use SHA-256-derived key:data:<hash> paths and Redis hash field normalization.
Credit accounting	Generation seeds starter credit. Additional credit consumption and webhook crediting are implemented; Stripe completed events are signature-checked and deduplicated.
Webhook resilience	Missing/invalid signatures return 400. Authenticated permanently malformed payloads are acknowledged to prevent retry storms, with operator logging.
Payload controls	Token calculator caps interactive input at 1 MB. Chat proxy caps request payloads at 512 KB. Log audit client parsing is capped at 10 MB.

External dependencies and required configuration

Dependency	Role
Vercel	Next.js deployment, middleware/proxy, Node serverless handlers, environment scopes, logs.
Upstash Redis	UPSTASH_REDIS_REST_URL and UPSTASH_REDIS_REST_TOKEN: key records, credit balances, rate limiting, webhook deduplication.
Stripe	STRIPE_SECRET_KEY and STRIPE_API_KEY_WEBHOOK_SECRET: prepaid credit checkout and verified webhook processing.
OpenAI	OPENAI_API_KEY: upstream only for the OpenAI-compatible Chat Completions proxy.
Node runtime	package.json declares Node >= 22.0.0 for local and deployment compatibility.

Privacy qualification

API key metadata can carry zero_data_retention: true and several surfaces avoid source-text persistence. This is an application control, not a blanket guarantee covering all external processors. Any enterprise data-retention commitment must additionally account for Vercel, Upstash, Stripe, OpenAI, and the customer configuration in scope.

6. Release readiness and next actions

Recommended production sequence

STEP	Action
1	Deploy the current Redis HGETALL normalization fix to production.
2	Generate a fresh key and verify POST /api/v1/keys/generate returns 201 and GET /api/v1/keys/balance returns 200 with Authorization: Bearer <key>.
3	Turn API_KEY_DIAGNOSTICS off and remove temporary header-level debug logging after the verification succeeds.
4	Run the release gates in the deployment branch: npx tsc --noEmit, npm run lint, npm test, and npm run build.
5	Use Stripe test-mode events to validate checkout metadata, one-time crediting, and malformed-payload operator alerts.
6	Run a staged end-to-end OpenAI proxy test with a non-sensitive prompt, observing credit consumption and upstream error passthrough.
7	Establish an editorial approval queue for generated research claims and maintain a source audit trail.

Risk register summary

Priority	Risk	Mitigation
High	Secrets or internal headers emitted by temporary diagnostics.	Disable diagnostics after validation; never paste raw platform headers or keys into tickets or chat.
High	Deployment environment drift across Vercel, Upstash, Stripe, and OpenAI.	Use scoped environment checklists and staging smoke tests before promotion.
Medium	Overstated scientific or efficiency outcomes.	Keep benchmark conditions, source citations, and accepted performance boundaries on every engagement.
Medium	Broad browser CORS policy for authenticated APIs.	Review production origin policy as SDK distribution and threat model mature.
Medium	Automated research graph expansion without editorial review.	Gate publication by provenance status and human source validation.

Conclusion

Maha Strategies has the core ingredients of a credible research-to-commercial infrastructure: structured research evidence, product-specific commercial surfaces, developer interfaces, middleware controls, and a metered operating model. The immediate priority is operational: complete the API-key production smoke test after redeploying the Redis normalization fix, then remove debug instrumentation and execute the full release gate.